

Dark Matter and Dark Energy: Two Names for One Screwed-Up Accounting System

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1. Introduction, The Double Mystery That Shouldn't Exist

Here is the situation modern cosmology finds itself in, stated plainly: roughly 27% of the universe's total energy budget consists of something called **dark matter**, invisible, detected only by its gravitational influence, with no confirmed direct detection despite decades of searching. Another 68% consists of something called **dark energy**, a mysterious repulsive force or vacuum energy that is causing the expansion of the universe to accelerate, detected only through redshift measurements and Type Ia supernova distance data. Add those two numbers together and you get 95%. That means the matter and energy we actually understand, protons, neutrons, electrons, photons, the entire Standard Model of particle physics, every atom in every star in every galaxy we've ever observed, accounts for only about 5% of what's actually out there.

Take a second to sit with that. The complete edifice of 20th-century physics, arguably the most successful intellectual project in human history, describes 5% of the universe. The other 95% gets two placeholder names and a collective shrug. In any other field, if your accounting system said 95% of your inventory was "unidentified," you wouldn't celebrate the discovery, you'd audit the accounting system. You'd ask whether the categories themselves were wrong.

That's exactly what we're doing here. This paper argues that dark matter and dark energy are not two separate, unsolved mysteries. They are two different observational signatures of the same underlying physical reality: the **substrate**, the actual physical medium that spacetime, fields, and particles are all expressions of. General Relativity (GR), which is the mathematical framework we use to describe gravity and cosmology, is geometrically brilliant. It correctly predicts the effects of dark matter and dark energy. But it cannot tell you what they physically

are, because GR was never designed to model the substrate directly. It treats spacetime as a geometric abstraction, a stage, rather than as a physical medium with mechanical properties. When it encounters phenomena that are, at their root, substrate phenomena, it can only describe them by their gravitational shadows. And because dark matter casts a different kind of gravitational shadow than dark energy does, GR sees two separate problems where there is actually one.

Unified Substrate Theory, UST, is a framework that takes seriously the idea that spacetime is a real physical medium. It has tension. It has density. It has pressure. It can carry waves. It has an internal mode structure, different ways it can oscillate and store energy. Once you adopt that framework, something remarkable happens: dark matter and dark energy stop being mysteries and start being expected. They're not anomalies in need of exotic explanations. They're the mechanical consequences of what happens when a physical medium undergoes an extreme compression-and-rebound event, what UST calls the **bounce**.

The argument unfolds as follows. Section 2 gives a brief introduction to UST's core concepts for readers who haven't encountered the framework before. Section 3 develops the UST account of dark matter in detail: it's non-electromagnetic substrate modes, primarily in the form of longitudinal compression wave fossils left over from the bounce. Section 4 develops the UST account of dark energy: it's the intrinsic tension and rebound dynamics of an expanding substrate medium. Section 5 explains how GR, despite being mathematically excellent, physically misidentifies both phenomena, and how its own formalism is responsible for splitting one reality into two artificial mysteries. Section 6 lays out testable predictions. Section 7 addresses the most serious objections. Section 8 zooms out to explain why getting this right matters enormously for the future of physics. Section 9 brings it all together.

We're not here to bash GR. GR is one of the greatest intellectual achievements in history. We're here to say that GR's geometric framework, as powerful as it is, has a vocabulary problem. And when your vocabulary doesn't have the right words for what you're seeing, you give things the wrong names. In this case, the wrong names are "dark matter" and "dark energy", and those

names have sent physics on a 50-year treasure hunt for things that don't exist in the form being searched for. Let's fix the accounting.

2. What the Substrate Is, A Brief UST Primer

Before we can explain what dark matter and dark energy actually are in UST's framework, we need to establish what the substrate is. If you're already familiar with UST's foundations, this section is a quick refresher. If you're coming in fresh, this is the conceptual foundation that everything else builds on. Don't skip it.

Start with what you probably already know about the quantum vacuum. In quantum field theory, the vacuum is not empty. It seethes with energy, virtual particle-antiparticle pairs constantly flickering in and out of existence, quantum fluctuations in every field, a ground-state energy that is very much real and measurable (the Casimir effect is one of the cleaner demonstrations of this). The vacuum, in modern physics, is already a deeply structured thing. It's not nothing. It never was nothing.

UST goes further. In GR and quantum field theory, the vacuum, however energetic, is still a backdrop. It's the stage on which physics happens. Fields live in spacetime. Particles are excitations of those fields. Spacetime geometry tells those fields and particles how to move (via the metric), and those fields and particles tell spacetime how to curve (via the stress-energy tensor). The relationship is deeply interactive, but there's still a conceptual separation: spacetime geometry is the container, and physics is the content.

UST collapses that separation. In UST, the **substrate** is not the stage, the substrate IS the physics. Fields, particles, and spacetime geometry are all expressions of the substrate's state. The substrate is the actual physical medium that everything is made of and made in. It has mechanical properties in the same sense that water has mechanical properties: it has a density, a tension, a pressure, a speed at which disturbances propagate through it, and a mode structure, a set of allowed patterns of oscillation and excitation.

The analogy I keep coming back to is this: in classical mechanics, you can describe the vibration of a drumhead purely geometrically, you track the displacement of every point on the surface as a function of time and position, and the math works perfectly. But if you want to understand *why* the drumhead vibrates the way it does, and *why* some frequencies are stable and others dissipate, you have to know that the drumhead is a physical medium with tension, mass density, and material properties. The geometry describes what happens. The physics of the medium explains why. GR is the geometry of the drumhead. UST is the physics of the medium.

2.1 Mode Structure, The Key Concept

The most important UST concept for this paper is **mode structure**. A mode, in physics, is a stable pattern of oscillation in a medium. When you pluck a guitar string, the energy doesn't just randomly jostle the string, it organizes into allowed vibrational patterns, the fundamental frequency and its harmonics. Each mode has a specific spatial pattern, a specific frequency, and a specific energy. The string doesn't have to vibrate in just one mode; it can simultaneously sustain energy in many modes, and the total vibration is the sum of all of them.

The substrate has a mode spectrum, a set of allowed excitation patterns determined by its tension, its density, its boundary conditions, and its history. This is not a metaphor or a loose analogy. It's the central physical claim of UST. The substrate is a real medium, it has a real mode structure, and the energy of the universe is distributed across that mode spectrum.

Now here's the critical point: some substrate modes couple to the electromagnetic field. When a substrate mode couples to EM, it means the excitation interacts with photons, it can emit light, absorb light, be detected by any instrument that relies on electromagnetic interaction. Those are the modes we call ordinary matter and radiation. Protons, neutrons, electrons, photons, these are all EM-coupled substrate modes. They're visible to us because they talk to light, and light is how we see everything.

But there is absolutely no physical requirement that ALL substrate modes couple to electromagnetism. In any real medium with a complex mode spectrum, the coupling of modes to external fields depends on the specific geometry and symmetry of each mode. Some modes

couple strongly. Some couple weakly. And some don't couple at all. The modes that don't couple to EM are what UST calls **non-EM substrate modes**. They have energy. They have mass-equivalent density. They curve spacetime, because anything with energy curves spacetime; that's as fundamental as it gets. But they emit no light, absorb no light, and are completely invisible to every photon-based detector we've ever built. They are, in the most literal and physically precise sense, dark.

2.2 The Bounce, Why the Mode Distribution Matters

UST situates our universe in a bouncing cosmology. The Big Bang was not a creation from nothing, a singularity popping into existence out of no prior state. That picture has always been philosophically troubling and mathematically ill-defined (GR itself breaks down at the singularity, which is a hint that the model is being pushed past its validity). In UST, the Big Bang was a **bounce**, a rebound from an extreme compression state. The universe before the bounce had been contracting, compressing the substrate to extraordinary densities and pressures, until the substrate's own internal pressure reversed the collapse and drove an explosive rebound outward.

This distinction matters enormously for understanding dark matter and dark energy. In a creation-from-nothing cosmology, you have to explain why the universe started with the particular energy distribution it did, why so much energy went into baryonic matter, why so much is "dark," why expansion has the particular dynamics it does. In UST's bouncing cosmology, you don't have to pull those answers from nowhere. The energy distribution of the post-bounce universe is the mechanical consequence of the bounce dynamics, which in turn depend on the pre-bounce state. The dark components aren't mysterious extras tacked on to explain observations. They're what you'd expect if you actually modeled what happens when you compress a real medium with a real mode structure to extreme density and then let it rebound. With that foundation in place, we can now talk about what dark matter actually is.

3. Dark Matter, The Non-EM Substrate Modes

3.1 The Mode Distribution Problem

Let's think carefully about what happens to energy when a complex physical medium undergoes a violent compression and rebound. Before the bounce, the substrate was compressed to extreme density. During the bounce, that compression energy was converted into kinetic energy of expansion, the rebound. As the substrate rebounded and expanded, that energy had to go somewhere. In the language of mode theory, the energy had to distribute itself across the substrate's available modes.

Now ask yourself: is there any physical reason why all of that rebound energy should go exclusively into EM-coupled modes? Should the bounce be perfectly selective, routing every joule of energy into modes that interact with photons and none into modes that don't? Think about what that would require. It would require that the EM-coupled modes have an overwhelming energetic advantage, that non-EM modes are either forbidden entirely or so unfavorable that they receive essentially zero energy from the bounce. There is no known physical mechanism that would guarantee this. In any real complex system, when you dump energy into a medium, it distributes across available modes weighted by the mode density (how many modes are available at each energy level) and the coupling strength (how easily energy flows into each mode). There is no reason to expect perfect selectivity toward EM-coupled modes.

In fact, it would be bizarre if all the energy went into EM-coupled modes. It would mean the bounce was cosmically lucky in a way that has no physical explanation. UST treats this not as luck but as a constraint: the bounce distributed energy across the full available mode spectrum. The EM-coupled fraction became baryonic matter and radiation, the stuff we can see. The non-EM fraction became dark matter, the stuff we can only detect gravitationally. The relative fractions are not free parameters to be tuned; they're consequences of the bounce dynamics and the substrate's mode density structure. The observed ratio (roughly 5 parts baryonic matter

to 27 parts dark matter, or about 1-to-5) is, in UST's framework, information about the bounce, a measurement of the substrate's mode density ratio at the time of the rebound.

This is one of UST's most important conceptual moves. It doesn't accept the 1-to-5 ratio as a mystery to be explained by particle physics. It takes that ratio as observational data about the substrate's mode structure. The question isn't "what particle is dark matter?" The question is "what does the 1-to-5 ratio tell us about the substrate's mode spectrum?" That's a completely different research program, and it points toward completely different observations.

3.2 Longitudinal Compression Modes, Fossils of the Bounce

Within the broad category of non-EM substrate modes, UST singles out one particular type as the primary component of what we observe as dark matter: **longitudinal compression modes**. To understand why, we need to distinguish between two fundamentally different kinds of waves in a medium.

Transverse waves oscillate perpendicular to their direction of propagation. Electromagnetic waves are transverse, the electric and magnetic fields oscillate at right angles to the direction the wave is traveling. Waves on the surface of water are also approximately transverse. Transverse waves require a medium with restoring forces that act perpendicular to the direction of propagation, in the case of EM waves in the substrate, this comes from the substrate's intrinsic tension structure and its coupling to the EM field.

Longitudinal waves, by contrast, oscillate parallel to their direction of propagation, compressions and rarefactions in the medium itself, like sound waves in air or pressure waves in a solid. Longitudinal waves are compression-mode phenomena. They require a medium with compressibility, with a well-defined relationship between local density and local pressure. The substrate, being a real physical medium with density and pressure, supports both types of waves. The EM-coupled modes are primarily transverse in character. The non-EM modes include a significant component of longitudinal compression structure.

Now think about what the bounce actually was: the most extreme compression event in the history of our universe. The substrate was compressed far beyond any density achieved in stellar interiors or even in neutron stars. When it rebounded, it didn't just smoothly expand, it launched longitudinal compression waves through itself. Think of it the way a bat hits a baseball: the impact creates a compression pulse that propagates through the ball. The substrate's bounce created compression pulses, longitudinal waves, that propagated outward through the expanding substrate.

In the immediate post-bounce era, these longitudinal compression modes existed across a vast spectrum of wavelengths, from the extremely short to the very long. What happened to them depended on their wavelength. The short-wavelength, high-energy longitudinal modes had enough energy to couple into the EM-interacting mode spectrum. They cascaded, through interactions in the hot dense post-bounce plasma, into the quarks and leptons and gauge bosons of the Standard Model. They became the baryonic matter and radiation of the hot Big Bang. This is the standard physics of the early universe, and it's entirely consistent with UST, UST just provides the "why" that standard cosmology doesn't have.

The long-wavelength longitudinal modes, however, were a different story. Their wavelengths were comparable to or larger than the coherence length of the substrate at the time of the bounce. They couldn't efficiently couple into the particle-mode spectrum, their wavelengths were simply too large to cascade into the short-scale structure of EM-coupled excitations. These long-wavelength longitudinal substrate compression modes persisted. As the universe expanded and cooled, they slowed their propagation (the substrate's effective wave speed changed as it expanded), and the longest of them effectively froze into quasi-static density perturbations, pressure variations in the substrate medium that are no longer rapidly propagating waves but slow-moving, large-scale concentrations of substrate mode energy.

These persistent long-wavelength longitudinal substrate compression structures are, in UST's account, the primary physical substance of dark matter. They are fossils of the bounce, the imprint of the initial compression event preserved in the large-scale structure of the substrate's mode distribution. They're not particles. They're not WIMPs (Weakly Interacting Massive

Particles). They're not axions. They're not sterile neutrinos. They don't need to be. They're what every physical medium leaves behind when it bounces: long-wavelength pressure modes that didn't have the energy to cascade into short-scale structure.

3.3 Why They Cluster the Way They Do

One of the central observational puzzles of dark matter is the galactic rotation curve problem. Let's be precise about what this means. In our solar system, the outer planets orbit more slowly than the inner ones, Mercury zips around the Sun in 88 days, Neptune takes 165 years. This makes sense: the Sun's gravity gets weaker with distance, so objects farther away don't need to move as fast to maintain a stable orbit. The same logic should apply to stars orbiting within a galaxy. Stars near the galaxy's center should orbit faster than stars at the edge. The gravity is weaker at the edge because you've left most of the galaxy's mass behind.

But that's not what we observe. When astronomers measure the orbital velocities of stars at different distances from the centers of spiral galaxies, they find that the velocities stay roughly constant, or even increase slightly, as you move outward. Stars at the edge of a galaxy orbit just as fast as stars near the center. Sometimes faster. The velocity curve, which should slope downward as you move outward (the way it does in the solar system), stays flat. This is called a "flat rotation curve," and it's been observed in hundreds of galaxies. It's not a measurement error. It's not a GR correction. It's a clean, robust observational fact.

The only way to explain flat rotation curves within Newtonian gravity and GR is to say that there is significantly more mass in and around galaxies than we can see, mass that extends far beyond the visible stellar disk, forming a large spherical "halo" around each galaxy. That extra mass is dark matter. Its gravitational pull keeps the outer stars moving fast even though they're far from the galaxy's luminous center.

In UST, this clustering behavior is natural. The non-EM substrate modes, including the longitudinal compression fossils, cluster around concentrations of baryonic matter because they follow the same substrate tension gradients that baryonic matter creates and responds to. Wherever normal matter is dense, it creates local tension concentrations in the substrate,

regions where the substrate is stressed more than its surroundings. These tension gradients act as wells that attract all substrate mode energy, both EM-coupled and non-EM. The dark matter halo doesn't have to be separately explained as a coincidence, it forms for exactly the same reason baryonic matter clumps: both types of mode energy respond to substrate tension gradients, and both contribute to those gradients.

There's also a scale consideration that explains why dark matter halos extend far beyond the baryonic disk. The longitudinal compression modes, being long-wavelength structures, have a characteristic spatial scale much larger than the scale on which baryonic matter collapses. Baryonic matter can radiate energy away electromagnetically (stars shine), losing angular momentum and collapsing into a thin disk. Non-EM modes can't do that, they have no way to shed energy through EM radiation. So they stay in the larger, more diffuse halo configuration, held there by the substrate tension gradient of the galaxy's total mass. The disk collapses; the halo doesn't. This is exactly the morphology we observe: a thin baryonic disk embedded in a large, roughly spherical dark matter halo.

There is one additional observational connection that deserves attention here. Stacy McGaugh and collaborators have compiled what's known as the SPARC dataset, a catalog of galaxy rotation curves with detailed measurements of baryonic matter distribution. What they found is striking: there is an extremely tight correlation between the observed gravitational acceleration in galaxies and the gravitational acceleration you'd predict from baryonic matter alone, across many orders of magnitude in galaxy mass and morphology. This "Radial Acceleration Relation" is hard to explain if dark matter is a separate population of particles that just happens to always distribute itself in exactly the right way relative to the baryonic matter. It's much more natural in a substrate mode framework, where both the baryonic matter and the dark matter (non-EM modes) are responding to the same underlying substrate tension gradients, they're correlated because they have the same source.

3.4 Why GR Can't See This

It's worth being precise about GR's limitation here, because it's not a failure of GR to do its job, it's a limitation of scope. GR's field equations say, in essence: the curvature of spacetime at any point is determined by the total mass-energy density and pressure at that point. That's it. GR doesn't ask what kind of mass-energy it is. It doesn't distinguish between a kilogram of protons and a kilogram of neutrinos and a kilogram of non-EM substrate mode energy. All mass-energy curves spacetime equally. That's the equivalence principle, and it's one of GR's deepest and best-tested results.

So GR knows dark matter is there. It can detect it through its gravitational effects, rotation curves, gravitational lensing, cluster dynamics, large-scale structure. GR's math correctly accounts for the gravitational influence of dark matter once you tell it how much mass to put in the halo. What GR cannot do is tell you what that mass is or why it doesn't interact with light. The geometry of spacetime encodes the presence of mass-energy but doesn't encode what type of mass-energy it is. Once you move from the geometry into the question of physical substance, GR goes silent.

That silence is where "dark matter" as a concept lives. It's not a physical substance GR has identified, it's a gap in GR's gravitational bookkeeping that GR itself can quantify but not explain. UST fills that gap not by adding a new particle to the Standard Model, but by recognizing that the substrate already has the mode structure to accommodate exactly this kind of dark mass-energy. The gap isn't a hole in nature. It's a hole in GR's vocabulary. UST provides the word.

4. Dark Energy, Container Tension and Rebound Acceleration

4.1 What Expansion Actually Is in UST

To understand UST's account of dark energy, we first need to be clear about GR's account of cosmic expansion, specifically, what GR does and does not say about why the universe expands.

In GR, cosmic expansion is described by the metric, the mathematical object that encodes distances between points in spacetime. When we say the universe is expanding, we mean the metric is changing over time: the distance between any two points that aren't gravitationally bound to each other is increasing. Crucially, GR specifies that this expansion is not motion through space. Galaxies are not flying apart through a pre-existing void. The space between them is literally growing. The metric itself is stretching. This is geometrically self-consistent and beautifully described by the Friedmann equations, which give you the expansion rate (the Hubble parameter) as a function of the universe's content, matter density, radiation density, and the cosmological constant.

What GR does not tell you is why the metric expands. The Friedmann equations tell you how the expansion rate depends on the energy content, but the energy content itself, especially the cosmological constant, is a given. You measure it; you plug it in; the equations run. The physical mechanism that drives expansion is absent from GR's description, because in GR, expansion is a geometric property of the spacetime manifold, not a mechanical property of a medium. There's nothing in GR's formalism that corresponds to a medium being pushed or pulled, there's only curvature evolving according to field equations.

UST has an actual physical account of expansion. The substrate is finite, not in the sense that you can reach a wall if you travel far enough in a straight line, but in the sense that it has a definite total mode-energy content and a definite topological boundary condition. The expansion of the universe is, in UST's terms, the relaxation of the substrate's boundary configuration following the bounce compression. Before the bounce, the substrate was compressed into an extreme-density state. The bounce reversed that compression. The post-bounce substrate has been expanding ever since, not because space itself is mysteriously doing something, but because the physical medium that constitutes spacetime was compressed and is now relaxing outward, carrying everything within it along as it goes.

This is a subtle but important distinction. In GR, asking "what is the expansion expanding into?" is a malformed question, there is no outside for it to expand into. In UST, the question has a partial answer: the substrate is relaxing into previously compressed configurations of

itself. The boundary condition changes as the substrate expands. This is not the same as saying there's an external void the universe expands into, it's saying the substrate's own internal configuration is evolving, and that evolution is what we call "expansion."

4.2 Container Expansion Tension, What Dark Energy Actually Is

The substrate, as a physical medium under tension, behaves in ways that have no analogue in GR's geometric description. One of those behaviors is the relationship between expansion and tension. When you stretch any medium, a rubber band, a spring, a fluid under negative pressure, you store energy in the stretch. That stored energy has mechanical consequences: it exerts forces, it drives subsequent motion, and it evolves as the medium's state changes.

As the post-bounce substrate expands, it does something that a simple elastic solid would not do. A simple elastic solid, when stretched, builds up restoring tension that eventually pulls it back to its rest configuration, that's how a spring works. But the substrate is not a simple elastic solid. It's a medium whose tension-to-density ratio evolves as it expands. At early times, when the substrate was dense (right after the bounce), the ratio of internal pressure to density was such that the kinetic energy of the rebound overwhelmed any restoring tension. The universe expanded rapidly and freely, this is the inflationary and early expansion epoch.

As the substrate expanded and its matter density dropped, the relative balance shifted. But here's the key insight that makes UST's account of dark energy different from simple elastic rebound: the topology of the substrate's boundary condition means that at large scales and late times, the substrate's tension does not act as a restoring force pulling things back together. Instead, it acts as a **container expansion tension**, a tension that, given the substrate's topological structure and the way the boundary condition scales with expansion, drives further expansion rather than resisting it.

This is counterintuitive, so let's work through the analogy carefully. Imagine a thin elastic membrane that is wrapped into a closed surface, a sphere, say. If you uniformly inflate it from the inside (increase the pressure), the tension in the membrane increases, and that tension acts to resist the inflation. Simple enough. Now imagine a different topology: a membrane that has

the property that as it expands outward, the tension vector at each point rotates to point outward as well, so that the tension that was a restoring force becomes a driving force above some critical expansion scale. This is not physically impossible; it's a consequence of the membrane's topology and material properties. UST proposes that the substrate's boundary topology has exactly this character: below some scale, the substrate tension resists expansion; above that scale (which is the current cosmological scale of our observable universe), the substrate tension drives further expansion.

That late-time outward substrate tension is **dark energy**. It's not a cosmological constant in the sense of a fixed, unchanging energy density tacked onto the field equations. It's not a new quantum field ("quintessence") invented specifically to explain the observations. It's the mechanical tension of an expanding substrate medium whose boundary topology drives accelerating expansion at late times. The "dark energy" we observe is the observational signature of the substrate doing what physical media do under tension: evolving dynamically in response to its own internal forces.

4.3 The Rebound Acceleration, Why Expansion Is Speeding Up

The discovery that the universe's expansion is accelerating, announced in 1998 based on observations of Type Ia supernovae by two independent teams, work that earned the 2011 Nobel Prize in Physics, was genuinely shocking. Before 1998, most cosmologists expected the expansion to be decelerating, because gravity should slow things down. Instead, the supernovae data showed that galaxies are receding from us faster today than they were in the past. The expansion is not just continuing; it's speeding up.

In GR's framework, explaining this acceleration requires introducing a term with negative pressure, a force that pushes outward rather than pulling inward. The simplest version is Einstein's cosmological constant, which he originally introduced (for entirely different reasons) and later called his greatest blunder. As a description of negative pressure, the cosmological constant works mathematically. As a physical explanation, it's nearly useless, because it gives you no mechanism, just a number. And when you try to compute what that number should be

from first principles using quantum field theory, you get a prediction that is off from the observed value by about 120 orders of magnitude. This is, without exaggeration, the worst theoretical prediction in the history of quantitative science.

UST offers a physical mechanism for the acceleration that doesn't require negative pressure as a mysterious given. The **rebound acceleration** in UST has a multi-phase structure. The initial bounce released kinetic energy into the substrate, that's what drove the early rapid expansion. As the expanding substrate cooled and the energy in high-frequency modes cascaded into particle formation (the processes of baryogenesis, nucleosynthesis, and structure formation), a significant fraction of the kinetic rebound energy was absorbed into organizing and structuring the matter content of the universe. This absorption of rebound energy into structure slowed the expansion, that's the decelerating expansion phase that standard cosmology correctly describes for the first several billion years after the bounce.

But the matter that formed from this process dilutes as the universe expands. As the universe gets bigger, the matter density drops, the same amount of matter spread through a larger volume. The rate at which matter density drops (as the inverse cube of the scale factor) is faster than the rate at which the substrate's expansion tension drops (which goes approximately as the inverse square of the scale factor, or more slowly, depending on the substrate's equation of state). So there comes a transition point, roughly 5-6 billion years after the bounce, consistent with the supernova observations, when the substrate's expansion tension overtakes the matter density in its influence on the expansion rate. After that transition, the substrate's own tension becomes the dominant driver of the expansion rate, and because that tension drives further expansion (as described in the previous section), the expansion accelerates.

Think of it this way. You fire a cannon (the bounce). The cannonball leaves the barrel fast (early rapid expansion). It slows as air resistance and gravity fight it (deceleration as matter density dominates). But suppose the cannonball has an internal spring that starts to extend only once the ball has cooled below a certain temperature. Once the ball has traveled far enough and cooled enough, the spring releases and accelerates the ball outward again. The spring energy was always there from the initial firing, it just took time and distance for the conditions to allow

it to release. UST's rebound acceleration is the cosmological equivalent of that spring: energy that was stored in the substrate's tension structure at the bounce, and that only becomes the dominant dynamical driver once the matter density has diluted sufficiently to let it take over.

This is not an ad hoc story. It has specific, testable consequences for how the dark energy density evolves over time, which we'll get to in Section 6.

4.4 Why GR Splits This Off as a "Mystery"

GR's handling of dark energy is a story of mathematical success masking physical ignorance. The Einstein field equations have a place for a cosmological constant, a term that acts like energy density in otherwise empty space, with the unusual property of having negative pressure. When you add this term to the equations and fit its value to the supernova data, you get a cosmological model (called Λ CDM, "Lambda-Cold Dark Matter") that fits essentially all large-scale cosmological observations quite well. The math works.

The problem is that GR provides no mechanism for the cosmological constant. It can't predict its value. It can't explain why it has the value it does. And when physicists tried to calculate what the vacuum energy density should be using quantum field theory, which should, in principle, contribute to the cosmological constant, they got a value roughly 10^{120} times larger than what's observed. One hundred and twenty orders of magnitude. Not a small discrepancy. Not a factor of two that could be explained by a missing correction. A discrepancy so enormous that it suggests either our quantum field theory calculation is catastrophically wrong, or our understanding of what the cosmological constant actually is is catastrophically wrong, or both.

GR's formalism gives you no leverage on this problem. In GR's framework, the cosmological constant is just a number, either it's what it is, or it isn't. You can postulate that quantum vacuum contributions cancel almost perfectly (a fine-tuning problem of absurd precision), or you can add a new dynamical field (quintessence, phantom dark energy, etc.) and add new free parameters to fit. Neither of these is a physical explanation. They're both mathematical

workarounds for the fact that GR has no physical model of what the vacuum energy actually comes from.

UST doesn't have this problem in the same form, because it doesn't treat the cosmological constant as a fundamental quantity. The effective dark energy density in UST is a consequence of the substrate's expansion tension and rebound dynamics, both of which are sourced by the bounce and evolve dynamically as the universe ages. It's not a constant. It's approximately constant over the redshift ranges we've measured so far, which is why Λ CDM fits the data. But it should deviate from constancy at higher precision and wider redshift ranges, a prediction we can test, and which DESI is already beginning to probe. More on that shortly.

5. How GR Got Two Mysteries from One Reality

5.1 GR's Vocabulary Problem

Let's be as precise as possible about what happened here, because it's a subtle failure mode that's worth understanding carefully. GR is not wrong. The Einstein field equations are not wrong. The Friedmann equations are not wrong. The predictions GR makes about gravitational lensing, orbital dynamics, black holes, gravitational waves, and the large-scale structure of the universe are extraordinarily accurate. GR is one of the two greatest theoretical achievements of 20th-century physics (quantum mechanics being the other), and it will still be one of the greatest theories ever constructed even if UST supersedes it in some domains.

The problem is scope, not correctness. GR was formulated in the language of differential geometry, curvature, metrics, geodesics, tensors. Its fundamental objects are the metric tensor (which encodes spacetime geometry) and the stress-energy tensor (which encodes the distribution of mass-energy and momentum). There is no slot in GR's formalism for "substrate mode," "longitudinal compression fossil," "EM coupling coefficient," or "boundary expansion tension." These concepts literally don't exist in GR's mathematical vocabulary. They're not wrong within GR's framework, they're undefined. They have no referent.

This means that when GR encounters phenomena that are, in UST's terms, substrate-level phenomena, it can only describe them through the one substrate property that GR does include: the contribution to the stress-energy tensor. Everything that affects spacetime curvature shows up in $T_{\mu\nu}$ (the stress-energy tensor), and that's all GR can see. A non-EM substrate mode? Shows up in $T_{\mu\nu}$ as energy density. Substrate expansion tension? Shows up in $T_{\mu\nu}$ as a term with negative pressure (or as the cosmological constant). Rebound kinetics in the substrate's geometry modes? Shows up as a time-varying contribution to the effective energy density.

GR flattens all of these physically distinct phenomena into the same mathematical object, $T_{\mu\nu}$, because $T_{\mu\nu}$ is the only window GR has into the physical world. Everything that can influence spacetime curvature goes in; the specific physical character of the source is invisible. This is analogous to trying to diagnose a patient's illness by measuring only their temperature. Temperature is a real measurement. It captures something real about the patient's state. But it flattens pneumonia, appendicitis, flu, and heat stroke all into the same readout: "elevated temperature." To distinguish them, you need a more detailed physical theory of what's actually going on inside the body.

5.2 The Splitting Mechanism

Here is exactly how one physical reality became two scientific mysteries. The process happened in two steps, separated by several decades.

First: in the 1930s and afterward, astronomers noticed that galaxies and galaxy clusters had more gravitational mass than their visible matter could account for. Fritz Zwicky observed it in galaxy clusters in 1933; Vera Rubin and Kent Ford documented it systematically in galactic rotation curves starting in the 1970s. The observational conclusion, that there was invisible mass providing extra gravitational pull, was interpreted within GR's framework as: there must be a new type of matter that doesn't interact with light. This became "dark matter," and the natural response within particle physics was to search for a particle that dark matter might be

made of. WIMPs, axions, sterile neutrinos, primordial black holes, all of these are particle-physics solutions to what is, in UST's account, a substrate-physics phenomenon.

Second: in 1998, the supernova observations revealed that the universe's expansion is accelerating. Within GR's framework, this required a term in the field equations with negative pressure, the cosmological constant or something similar. This became "dark energy." Because it lived in a different part of GR's field equations (the left-hand side, as a modification to the geometry, or equivalently as a vacuum energy density contribution to the right-hand side), it was treated as a completely separate problem from dark matter. Two different terms in the equations, two different research communities, two Nobel Prizes, two separate major surveys (one optimized for dark matter, one for dark energy).

But both the "extra gravitational mass" and the "accelerating expansion" are, in UST's account, properties of the same physical object: the substrate in its post-bounce configuration. The extra gravitational mass is the energy stored in non-EM substrate modes. The accelerating expansion is the substrate's own boundary tension and rebound dynamics manifesting at cosmological scales. They're both the substrate. GR's formalism put them in different places in the field equations, one in $T_{\mu\nu}$ on the matter side, one as the cosmological constant on the geometry side, and that formal separation made them look like two separate physical phenomena. They're not. It's one medium with two different observable properties.

This is the accounting error. Not a numerical error. A categorical error. It's the equivalent of a mechanic who doesn't know cars very well looking at an engine and saying "I see a problem with the electrical system and a completely separate problem with the fuel system," when the actual problem is a single faulty fuel pump that's draining the battery by running continuously. The electrical symptom and the fuel symptom are real. But they have one cause, and understanding them separately prevents you from finding it.

5.3 What Unification Looks Like in UST

It's important to be precise about what "unification" means here, because physics has a specific and demanding standard for this term. Unification doesn't mean "we say it's all one thing and

now we don't have to explain anything." It means identifying a single physical object or framework from which both phenomena follow as natural consequences, in a way that reduces the number of free parameters, makes new testable predictions, and provides physical mechanisms where previously there were only mathematical descriptions.

Maxwell's unification of electricity and magnetism is the canonical example. Before Maxwell, electricity and magnetism were studied separately. They had separate laws, separate constants, separate experimental traditions. Maxwell showed that they were both aspects of a single electromagnetic field, one physical object with two different manifestations depending on the reference frame of the observer. This unification wasn't just aesthetically satisfying. It made new predictions (electromagnetic waves, and their speed, which turned out to equal the speed of light), it reduced the number of independent constants (the electric constant and magnetic constant were related through the speed of light), and it opened entirely new domains of physics.

UST's unification of dark matter and dark energy aspires to the same standard. The single physical object is the substrate. Dark matter is the non-EM mode energy in the substrate, the energy stored in modes that don't couple to photons, primarily the longitudinal compression fossils of the bounce. Dark energy is the substrate's boundary expansion tension and rebound dynamics, the mechanical consequences of the bounce acting on the substrate's geometry at late times. Both phenomena follow from the same event (the bounce), operating through the same medium (the substrate), governed by the same physical principles (the mechanics of a real medium with mode structure, tension, density, and pressure).

The unification makes new predictions, which we'll discuss in Section 6. It reduces the number of independent mysteries from two to zero, by showing that both are expected consequences of the bounce rather than unexplained inputs. And it provides physical mechanisms: the dark matter mechanism is mode-energy distribution in a medium with a specific mode spectrum; the dark energy mechanism is boundary tension and rebound kinetics in that same medium.

These aren't two separate things that happen to have a common origin story. They're two different observable aspects of one medium's state. The substrate after the bounce is the dark matter and the dark energy, in the same way that a stretched guitar string is simultaneously a source of mechanical tension and a carrier of specific vibrational modes. You can describe them separately, and sometimes it's useful to, but they're the same physical object, and you can't understand either one completely without understanding both.

6. Observational Predictions and Tests

A physical theory that makes no testable predictions beyond those that motivated it isn't much of a theory. UST makes several specific predictions that differ from the standard Λ CDM cosmological model. Some of these are already being tested by current and near-future experiments. Let's go through them clearly.

6.1 Dark Matter Predictions

The most direct prediction is negative: UST predicts that dark matter will not be detected as a particle in any experiment that looks for particle-physics interactions. This includes direct detection experiments like LUX, PandaX, and XENONnT (which look for WIMPs scattering off atomic nuclei deep underground), collider experiments like the LHC's searches for dark matter particles in proton-proton collisions, and indirect detection experiments that look for gamma rays or cosmic rays from dark matter annihilation. These experiments have now searched for WIMPs across a large fraction of the theoretically motivated parameter space with no confirmed detections. UST's interpretation of those null results is simple: they're looking for the wrong thing. Dark matter is not a particle. There is no dark matter particle to find. The non-EM substrate modes that constitute dark matter don't interact with Standard Model particles in the way a WIMP would. The null results are expected, and they will continue.

This is a strong prediction. If a convincing dark matter particle signal shows up in a direct detection experiment, that's a serious problem for UST's account. But the track record so far is consistent: decades of increasingly sensitive experiments, no detection, with Λ CDM's

theoretically preferred WIMP territory largely ruled out. UST says keep going and keep finding nothing, because you will.

On the positive side, UST predicts that dark matter's distribution should reflect the large-scale mode structure of the substrate, which in the post-bounce universe should closely trace the cosmic web, the large-scale filamentary network of dark matter and galaxies that we observe. This is consistent with what we see: dark matter doesn't appear as randomly distributed point masses. It forms halos around galaxies and clusters, connects those halos with filaments, and avoids the large empty voids between filaments. This filamentary topology is exactly what you'd expect from standing-wave and near-standing-wave patterns in the substrate's mode structure, large-scale coherence imposed by the initial bounce conditions.

UST also predicts a specific internal structure for dark matter halos: they should have solitonic core profiles rather than the "cuspy" NFW (Navarro-Frenk-White) profiles predicted by cold dark matter simulations. A **solitonic halo structure** is a stable, non-dispersing wave concentration, the kind of structure you get in non-linear wave media where self-gravitation prevents dispersal. The outermost halo follows a standard density falloff, but the core is dominated by a solitonic concentration of substrate mode energy. Observations of dwarf galaxies, the smallest gravitationally bound dark matter structures, show a tension with NFW profiles (the "core-cusp problem"). Solitonic cores are a competitive explanation and are being actively tested with current observational data. UST naturally predicts them.

Finally, the tight correlation between baryonic matter distribution and total gravitational acceleration (the Radial Acceleration Relation from the SPARC data) is a natural consequence of the UST framework, where both baryonic and non-EM mode energy respond to the same substrate tension gradients. Λ CDM has difficulty explaining this tightness without additional tuning. UST gets it for free.

6.2 Dark Energy Predictions

The most important UST prediction about dark energy is that it should not be a true cosmological constant. The substrate's expansion tension and rebound dynamics evolve as the

universe expands. This means the effective dark energy density should vary with time, slowly, but measurably, especially across wide redshift ranges.

This is precisely the question the Dark Energy Spectroscopic Instrument (DESI) collaboration is investigating right now. DESI has measured the redshifts of more than 30 million galaxies and quasars and analyzed Lyman-alpha forest spectra from over 820,000 quasars. Their results, released in 2024 and refined in 2025, suggest that dark energy may indeed be time-varying. When they parameterize the dark energy equation of state using the standard w_0 - w_a framework (where w_0 is the present-day value and w_a characterizes the time evolution), they find values inconsistent with a pure cosmological constant (which would have $w_0 = -1$ and $w_a = 0$) at significance levels between 2.8σ and 4.2σ , depending on which supernova dataset is combined with the DESI data. The current best-fit values cluster around $w_0 \approx -0.75$ to -0.84 and $w_a \approx -0.62$ to -1.09 , a dark energy that was weaker in the past and is strengthening over time.

Let that sink in for a moment. If DESI's results hold up, and they'll be tested against data from the Euclid satellite and the Vera Rubin Observatory's Legacy Survey of Space and Time (LSST) in the coming years, then the cosmological constant is not constant. That's a direct falsification of the simplest dark energy model and a strong signal that there is genuine dynamics in whatever is driving the accelerating expansion. UST's substrate expansion tension model naturally produces a time-varying effective equation of state, because the substrate's tension dynamics evolve as the universe expands. The DESI results are, at minimum, pointing in the direction that UST predicts.

UST also makes a more ambitious prediction: there should be a specific relationship between the conditions of the bounce (which are encoded in early-universe observables, particularly the spectrum of primordial gravitational waves detectable in CMB B-mode polarization) and the current dark energy density. In Λ CDM, the cosmological constant is a free parameter, you measure it, and that's its value, with no deeper explanation. In UST, the dark energy is sourced by the bounce, so the bounce's energy scale should be related to the observed dark energy density through the substrate's dynamics. If we can independently constrain the bounce energy scale (through primordial gravitational wave observations by experiments like CMB-S4 or

LiteBIRD), we can test whether the implied dark energy density is consistent with the observed value. This is a cross-check that Λ CDM simply cannot make.

6.3 The Bounce Signature

More broadly, UST's entire account of dark matter and dark energy rests on the bounce being a real physical event that left detectable imprints. This means UST predicts the existence of pre-bounce signatures in the CMB, particularly in the spectrum of primordial gravitational waves at scales corresponding to the mode structure of the pre-bounce universe. Standard inflationary cosmology also predicts a gravitational wave background, but with a specific spectral shape. UST's bounce cosmology predicts modifications to that spectral shape that, in principle, could be distinguished. If CMB-S4 or a future space-based gravitational wave observatory detects a gravitational wave background with spectral features inconsistent with standard inflation but consistent with a bounce-rebound scenario, that's a major point for UST's framework.

The predictions are specific, they're falsifiable, and several of the relevant experiments are either running now or will be running within the next decade. That's exactly where a new theoretical framework should be: making clear, testable claims that current and near-future observations can evaluate.

7. Common Objections and Responses

Any framework that challenges the standard account of dark matter and dark energy is going to face serious objections, and it should. Let's go through the most serious ones without softening the answers.

7.1 "UST Is Just Rebranding the Vacuum Energy Problem, Not Solving It"

This is probably the sharpest objection. The claim goes: you're calling dark energy "substrate expansion tension" instead of "vacuum energy," but haven't you just pushed the mystery back one level? Now you have to explain what determines the substrate tension, what sets the

bounce energy scale, and why the rebound dynamics produce the dark energy density we observe. Isn't that just as mysterious as a cosmological constant?

The answer is: no, and here's why. The cosmological constant problem in Λ CDM isn't just "we don't know the value", it's "the value we observe is 120 orders of magnitude smaller than what quantum field theory predicts, with no known mechanism for the cancellation." That 120-order-of-magnitude discrepancy isn't a gap in our knowledge that more calculations will fill; it's a sign that something fundamental is wrong with how we're computing the vacuum energy contribution. UST sidesteps this problem because it doesn't try to compute dark energy from quantum vacuum fluctuations in the first place. In UST, the dark energy isn't the vacuum energy, it's the substrate's mechanical tension from the bounce. These are different physical things, and they don't add together in the way Λ CDM's cosmological constant requires.

Yes, UST needs to explain what determines the bounce energy scale and how the substrate's tension dynamics produce the specific dark energy density we observe. That's a legitimate outstanding problem, and the full quantitative treatment hasn't been worked out. But the outstanding problem is: "derive the dark energy density from the bounce physics." That's hard, but it's a well-posed physics problem with a specific mechanism to compute from. The outstanding problem in Λ CDM is: "explain why 120 orders of magnitude of vacuum energy precisely cancel, leaving only the tiny observed residual." That's not a computation problem, it's a sign of a framework failure. UST's outstanding problem is harder; Λ CDM's outstanding problem is arguably insoluble within its own framework.

7.2 "Where's the Math?"

This is fair, and it deserves a straight answer. The full mathematical formalism of UST, specifically, the quantitative model of substrate mode dynamics, the derivation of the mode spectrum from first principles, the calculation of the post-bounce energy distribution across EM-coupled and non-EM modes, and the dynamical equation governing substrate expansion tension, is still being developed. What exists now is a well-specified conceptual framework with

clear physical mechanisms. This paper is a conceptual account. It is not a finished mathematical theory.

That's not a fatal flaw, it's a stage of development. Einstein's special relativity started as a conceptual account of what would happen if the laws of physics were the same in all inertial frames. The mathematics followed. GR started as a conceptual account of the equivalence principle and the idea that gravity is curvature. The field equations came later, after years of work. Conceptual clarity is not optional extra garnish on top of real physics, it's the prerequisite for knowing which mathematics to develop. UST is at the stage where the conceptual framework is being specified and sharpened. The mathematical work is next.

What the mathematical formalism needs to produce: (1) a dynamical equation for substrate mode energy distribution as a function of time and scale, from which dark matter density profiles can be derived; (2) a substrate expansion equation analogous to the Friedmann equations but including boundary tension terms, from which the dark energy equation of state can be computed; and (3) quantitative predictions for the bounce energy scale and its relationship to observable quantities. These are hard problems. They're tractable problems. The path forward is clear, even if the path hasn't been fully walked yet.

7.3 "Substrate Modes Can't Form Stable Halos, They'd Disperse"

This is a substantive physics objection. Waves in a medium propagate and disperse. Why would longitudinal substrate compression modes clump into stable halos around galaxies rather than just propagating away? If dark matter is waves in the substrate, shouldn't those waves spread out and smooth away over cosmological time?

The answer has two parts. First, the long-wavelength longitudinal modes we're talking about aren't freely propagating waves in the ordinary sense. Their propagation speed in the late-universe substrate is extremely slow compared to the age and scale of the universe, they're effectively quasi-static density perturbations in the substrate, not rapidly traveling waves. Their "wavelength" corresponds to galactic or super-galactic scales, and their "period" of oscillation

would be longer than the current age of the universe. For practical purposes, they're density concentrations in the substrate, not oscillating waves.

Second, and more importantly, these modes are self-gravitating. They have energy, energy gravitates, and their own gravitational self-interaction can stabilize them against dispersal. In non-linear wave physics, this is well understood: when wave energy is large enough relative to the medium's restoring forces, and when there is an attractive self-interaction (like gravity), waves can form stable, non-dispersing solitonic structures. Solitons in condensed matter physics are the canonical example, stable, localized wave concentrations that propagate without spreading because the non-linearity of the medium exactly compensates the dispersal tendency of the wave.

UST's proposal is that dark matter halos are gravitationally self-bound solitonic structures in the substrate's mode energy. This is not an ad hoc add-on, it's the expected behavior of self-gravitating waves in a non-linear medium. It has a specific observational signature: the density profile of the halo's core should follow a solitonic shape (flatter in the core, transitioning to a steeper outer profile), rather than the NFW cusp. As noted in Section 6, this prediction is actively being tested against observations of dwarf galaxies, where the core-cusp discrepancy is most acute. Early results are consistent with solitonic cores. The prediction is falsifiable and currently unrefuted.

7.4 "Doesn't This Conflict with the Bullet Cluster?"

The Bullet Cluster is one of the strongest pieces of observational evidence for the existence of dark matter as a separate component from baryonic matter. It's a pair of galaxy clusters that collided about 150 million years ago. During the collision, the hot gas of each cluster (detected in X-rays) was slowed by electromagnetic interactions, the gas clouds collided and partially shocked. But the gravitational mass of each cluster, measured by the way the clusters' gravity bends background light (gravitational lensing), passed right through the collision zone without slowing down. The lensing mass is offset from the X-ray gas, clearly separated. This is taken as evidence that most of the mass in each cluster is in a component that doesn't interact

electromagnetically, dark matter, because only something that doesn't interact with EM fields would pass through the other cluster unimpeded.

Some people raise this as a problem for wave-based dark matter models, on the grounds that waves might interact with each other or with the baryonic matter in some way. But it's actually not a problem for UST at all, it's exactly what UST predicts. The non-EM substrate modes don't couple to the electromagnetic field. They don't interact with the baryonic matter of the clusters electromagnetically. When two clusters of non-EM substrate modes pass through each other, they experience no electromagnetic drag. They pass through because they are, by definition, electromagnetically decoupled from the baryonic matter and from each other's EM-coupled components. The gas slows; the substrate modes don't.

The only UST requirement for the Bullet Cluster consistency is that non-EM substrate modes must not interact strongly with each other through the substrate in a way that would cause momentum transfer during the collision, essentially, that the modes are approximately collisionless on the timescale of the cluster collision. UST is consistent with this: the long-wavelength longitudinal modes interact primarily through gravity (long-range) rather than through short-range substrate scattering (which would be relevant only for very short-wavelength, high-frequency modes). For the large-scale mode concentrations that constitute dark matter halos, effective collisionlessness is a natural property of the long-wavelength limit. The Bullet Cluster is not a problem for UST; it's a confirmation of the key property, electromagnetic decoupling, that UST's account of dark matter requires.

8. The Bigger Picture, Why This Matters Beyond Cosmology

8.1 The 95% Problem

Let's be direct about what it means to have 95% of the universe's energy content in the "unknown" category. Every simulation of cosmic structure formation, every calculation of how galaxies form, how clusters grow, how the cosmic web develops, is built on a model of dark matter and dark energy. Those models are currently parameterized versions of things we don't

physically understand. The simulations are sophisticated and impressive, and they reproduce many observed features of the large-scale universe. But they're sophisticated parameterizations of ignorance. They work because we tune the parameters to match observations, not because we understand the physical mechanisms well enough to derive the parameters from first principles.

This matters because when something unexpected shows up in observations, and things do keep showing up, from the Hubble tension to the apparent excess of massive galaxies at high redshift seen by JWST to the DESI dark energy anomaly, we have no physical framework robust enough to know whether it's a genuine anomaly, a measurement artifact, or a signal of new physics. When you're 95% confused about your foundations, your ability to interpret anomalies is severely limited. Getting the physics of dark matter and dark energy right is not a specialized puzzle for cosmologists, it's the prerequisite for having a reliable foundation for all of cosmology and for a significant portion of fundamental physics.

8.2 The Bouncing Cosmology Implication

If UST is right about dark matter and dark energy, if both are substrate phenomena sourced by the bounce, then both of the biggest cosmological mysteries in modern physics are telling us the same thing: the universe did not begin from nothing. There was a prior state. There was a compression. There was a rebound. The signatures of that event are written into both the dark matter distribution (the mode fossils of the bounce) and the dark energy dynamics (the rebound tension of the substrate). We're not just speculating about a cyclical cosmology for philosophical reasons. We're saying that the data we already have, the rotation curves, the large-scale structure, the dark energy density, its potential time variation, are all fossil records of the bounce, and reading them correctly tells us that cosmology before the Big Bang is a real subject with real observational handles.

This is a profound shift in perspective. The standard cosmological model treats the Big Bang as the beginning, not just the beginning of time we have access to, but effectively a wall beyond which no physics applies. UST treats it as a transition, the most violent transition in our

universe's history, but a transition nonetheless, from a prior compressed state to our current expanding state. If that's right, then the question "what happened before the Big Bang?" is not a philosophical dead end. It's a physics question with potentially observable consequences. The pre-bounce conditions set the initial mode spectrum; that mode spectrum determines the dark matter distribution; the dark matter distribution is something we can measure. The chain of inference runs all the way from current observations back through the bounce to the pre-bounce state. That's extraordinary, if UST's framework is right.

8.3 Redirecting the Search

There is a practical consequence that deserves to be stated plainly. The search for particle dark matter has consumed enormous resources, intellectual, financial, and institutional, for several decades. The LEP collider searched for WIMP candidates at the electroweak scale. The LHC has performed extensive dark matter searches. Dozens of underground direct detection experiments have been built, operated, and successively scaled up. Axion searches are ongoing. Sterile neutrino searches are ongoing. These are not wasted efforts, the experiments have advanced detector technology, constrained particle physics models, and taught us a great deal. But if UST is right, and dark matter is non-EM substrate modes rather than any kind of Standard Model extension, then the particle dark matter search program will continue to find nothing, at any energy scale and with any detector technology, forever. Because the thing being searched for doesn't exist in the form being searched for.

Knowing this, or even taking it seriously enough to pursue UST's alternative seriously, would redirect resources toward the right tools. Gravitational wave observatories (LIGO, LISA, and future detectors) are tools for probing the substrate directly, since gravitational waves are substrate disturbances. CMB analysis for bounce signatures, particularly B-mode polarization at large angular scales, is a probe of the pre-bounce mode structure. Precision measurements of the dark energy equation of state across wide redshift ranges (exactly what DESI, Euclid, and Rubin are doing) are measurements of the substrate's expansion tension dynamics. These are the right experiments for testing UST's framework. Some of them are already running. We don't

need to build new instruments from scratch, we need to ask different questions of the instruments we already have.

9. Conclusion, Put It Back Together

Here's what we've argued, section by section, and what it adds up to. GR is a brilliant description of how mass-energy affects spacetime geometry. It is so good that it can detect dark matter and dark energy through their gravitational effects even without having any idea what they physically are. The rotation curves, the gravitational lensing maps, the supernova distance data, GR processes all of it correctly and tells you that something is there. But GR's mathematical success at quantifying these phenomena has, paradoxically, obscured the physical question of what they actually are. Because GR can describe the effects without needing a physical model of the cause, and because GR's formalism puts the two effects in different places (dark matter in the stress-energy tensor, dark energy as the cosmological constant), physics has been treating them as two separate, unrelated mysteries for decades.

They're not separate. They're not unrelated. They're not mysterious in the sense of being fundamentally beyond explanation. They are both properties of the substrate, the physical medium that spacetime, fields, and particles are all expressions of, in its post-bounce configuration.

Dark matter is the non-EM mode energy of the substrate. Most of it is in the form of long-wavelength longitudinal compression modes, fossil pressure waves from the bounce that didn't have enough energy to cascade into the EM-coupled mode spectrum and become baryonic matter. These modes have energy, they gravitationally cluster, they form stable solitonic halos around galaxies, and they don't interact with photons. They're dark not because they're exotic, but because they are substrate excitations in modes that simply don't couple to the electromagnetic field. They're invisible for the same reason that radio waves are invisible to a photographic plate, wrong coupling, not wrong existence.

Dark energy is the substrate's boundary expansion tension and rebound dynamics, the mechanical consequences of the bounce playing out at cosmological scales over cosmological time. The early rapid expansion was the direct kinetic rebound from the bounce. The subsequent deceleration was the absorption of kinetic energy into matter formation. The current accelerating expansion is the re-emergence of the substrate's own tension-driven dynamics as the matter density dilutes and the substrate's intrinsic expansion tension becomes the dominant driver. These aren't three separate phases requiring three separate explanations, they're one medium going through one physical process: bouncing, expanding, and evolving dynamically as its content dilutes and its tension structure expresses itself at progressively larger scales.

GR split these into two mysteries because GR's formalism had no place for the physical concepts that would unify them. There's no slot in the Einstein field equations for "substrate mode type," "EM coupling coefficient," or "boundary expansion tension." So GR described the gravitational shadows of both phenomena, accurately, and gave each shadow a name. The names stuck. The research programs organized around the names. The mystery deepened as the names were taken to be real categories rather than observational labels for aspects of something more fundamental.

UST's answer is not to abolish GR, GR's math is right and will remain useful for all the things it's always been useful for. The answer is to situate GR correctly: as a theory of spacetime geometry that operates within a deeper physical reality, the substrate, which GR's formalism approximates but does not fully describe. In the same way that thermodynamics operates correctly within statistical mechanics (thermodynamic laws are real and useful, but their physical basis is statistical mechanics of atoms), GR operates correctly within UST's substrate framework, but the physical basis for what GR describes as "mass-energy curving spacetime" is the substrate's mode structure and dynamics.

The path forward is clear. Develop the mathematical machinery to make the qualitative mechanisms in this paper quantitative. Design the observations, or reinterpret the observations already in progress, to test the specific predictions: no dark matter particle, time-varying dark

energy, solitonic halo profiles, tight coupling between dark matter and baryonic matter distributions through shared substrate tension gradients, and bounce signatures in the primordial gravitational wave background. Build toward a full dynamical theory of the substrate that gives you GR as a geometric approximation and gives you UST's mode dynamics as the deeper physical layer.

We haven't solved everything. The full quantitative theory is work that still needs to be done, and that work is hard. But we've done something that matters as much as solving the details: we've stopped asking the wrong questions. "What particle is dark matter?" is the wrong question. "What drives dark energy?", when posed as a search for a new field, is the wrong question. The right questions are: "What is the substrate's mode spectrum, and how did the bounce distribute energy across it?" and "What are the substrate's expansion tension dynamics, and how do they evolve?" Those questions have physical mechanisms behind them, and physical mechanisms can be calculated, tested, and refined.

That's where progress starts. Not with a mystery, but with a mechanism. Not with two names for two unknown things, but with one medium doing one thing, seen from different angles, at different scales, through different instruments. It's all substrate. It was always substrate. And once you see that, the 95% problem stops being a confession of ignorance and starts being a research program.

Key Terms Glossary

Term	Definition
Substrate	In UST, the actual physical medium that spacetime geometry, fields, and particles are all expressions of. Unlike GR's spacetime (a geometric abstraction), the substrate is a real physical medium with measurable properties: tension, density, pressure, and a mode structure. Everything in the universe is either a substrate excitation or a property of the substrate's state.
Non-EM Substrate Modes	Excitation patterns of the substrate that do not couple to the electromagnetic field. They carry energy (and thus have mass-equivalent density and curve spacetime), but they do not emit or absorb photons. They are invisible to all photon-based detectors. In UST, these modes are the physical substance of what cosmology calls "dark matter."
Longitudinal Compression Modes	Substrate excitations in which the medium oscillates (or is structured) parallel to the direction of propagation, compression and rarefaction waves in the substrate itself, analogous to sound waves in air. Contrasted with transverse modes (where oscillation is perpendicular to propagation), which characterize electromagnetic radiation. UST proposes that long-wavelength longitudinal compression modes, formed during the bounce rebound and persisting as quasi-static density structures, are the primary component of dark matter.
Bounce	In UST's cosmological framework, the event corresponding to the "Big Bang", not a creation from nothing, but a rebound from an extreme compression state of the substrate. The pre-bounce universe had been contracting; the substrate's internal pressure reversed the collapse and drove the explosive expansion we observe as the aftermath. The bounce distributed energy across the substrate's full mode spectrum, producing both the baryonic matter (EM-coupled modes) and dark matter (non-EM modes) of our universe.
Rebound Acceleration	The multi-phase kinetic dynamics of the substrate's post-bounce expansion. The initial rapid expansion was direct kinetic rebound energy from the bounce. A deceleration phase followed as that kinetic energy was absorbed into matter formation. The current accelerating expansion phase, what cosmology calls the effect of "dark energy", is the substrate's own intrinsic expansion dynamics re-emerging as the dominant driver once matter density dilutes sufficiently. The energy driving the current acceleration was stored in the substrate from the bounce; it simply took cosmological timescales to manifest.
Container Expansion Tension	The intrinsic mechanical tension of the substrate as a physical medium with a finite topological boundary condition. As the post-bounce substrate expands, its boundary condition evolves in a way that, at large scales and late cosmological times, drives further outward expansion rather than resisting it. This substrate tension is UST's physical account of what cosmology calls "dark energy", not a vacuum energy constant or a new field, but the mechanical behavior of the substrate's boundary under expansion.

Term	Definition
Solitonic Halo Structure	A gravitationally self-bound, non-dispersing concentration of substrate mode energy that forms the core of dark matter halos in UST's framework. Unlike freely propagating waves, which disperse over time, solitons are stable wave concentrations where the tendency to disperse is exactly compensated by the non-linear self-gravitational attraction of the mode energy. UST predicts that dark matter halo cores should have solitonic density profiles, flatter than the "cuspy" NFW profiles predicted by cold particle dark matter models, a prediction currently being tested against observations of dwarf galaxies.
ΛCDM	Lambda-Cold Dark Matter, the standard cosmological model. "Lambda" refers to the cosmological constant (Einstein's symbol for it), which accounts for dark energy in this model as a fixed, unchanging energy density of empty space. "Cold Dark Matter" refers to the hypothesis that dark matter consists of slow-moving (cold) massive particles. Λ CDM fits current observational data well as a parameterization but provides no physical mechanisms for what dark matter particles are or why the cosmological constant has the value it does.
Equation of State (w)	In cosmology, the ratio of a fluid's pressure to its energy density, used to characterize how different components of the universe affect its expansion. Normal matter has $w \approx 0$ (pressure-free). Radiation has $w = 1/3$. A cosmological constant has $w = -1$ (negative pressure). The parameter w for dark energy is currently being measured by DESI and other experiments. A value of $w = -1$ exactly corresponds to a true cosmological constant; deviations indicate dynamic dark energy. UST predicts measurable deviations from $w = -1$ because the substrate's expansion tension is dynamic, not constant.